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18 December 1978
SUPERSEDING
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31 August 1964

MILITARY SPECIFICATION

BURNER, GAS, LABORATORY, BUNSEN, AUTOMATIC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

* 1.1 Scope. This specification covers an automatic Bunsen burner, adjustable for use with manufactured, natural, mixed (natural and manufactured) gas, or bottle liquid petroleum gas such as propane, butane, etc.

2. APPLICABLE DOCUMENTS

* 2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-585	- Boxes, Wood, Wirebound.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	- Boxes, Fiberboard.
PPP-B-676	- Boxes, Setup.

MILITARY

MIL-L-10547	- Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Headquarters, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, Code ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6640

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Material. Material shall be as hereinafter specified, and shall be suitable for the purpose intended.

* 3.2 Construction. The burner shall be so constructed that the pilot shall burn continuously and the main burner shall be activated by depression of the "on-off" platform. The burner shall be adjustable for use with manufactured, natural, mixed (natural and manufactured) gas, or bottled liquid petroleum gas. The burner shall consist of the following components: base housing, valve actuating housing, "on-off" platform, detachable burner, draft guard, detachable nozzle for each gas, and waxing cup. Shall be in accordance with figure 1 for style and design.

3.2.1 Base housing. The base housing shall be fabricated of zinc alloy, and shall be "D" shaped and die cast. The housing shall contain a needle valve with an exterior nozzle orifice at one end and a knob at the other end for regulating the volume of gas for the desired flame, and shall have a serrated gas inlet connection to accommodate rubber tubing, 1/4 to 5/16 inch in inside diameter. Base housing shall also have a removable metal valve mechanism to which shall be affixed a neoprene seal, the valve being held normally closed against the fixed metal portion of the base housing by a bronze spring.

3.2.2 Valve actuating housing. The valve actuating housing shall be fabricated of zinc alloy and shall be die cast. The housing shall be mounted onto the base housing, and shall contain the valve actuating mechanism. A nylon neoprene-impregnated diaphragm shall be fitted between the housings, and shall permit opening and closing of the valve mechanism without leakage of gas to the atmosphere. The housings shall be securely joined by four screws.

* 3.2.3 "On-Off" platform. The "on-off" platform shall be fabricated of zinc alloy and shall be die cast. Platform shall be flat and circular, shall be mounted on top of the actuating housing, and shall be attached to the valve actuating mechanism. Depressing and rotating the platform in

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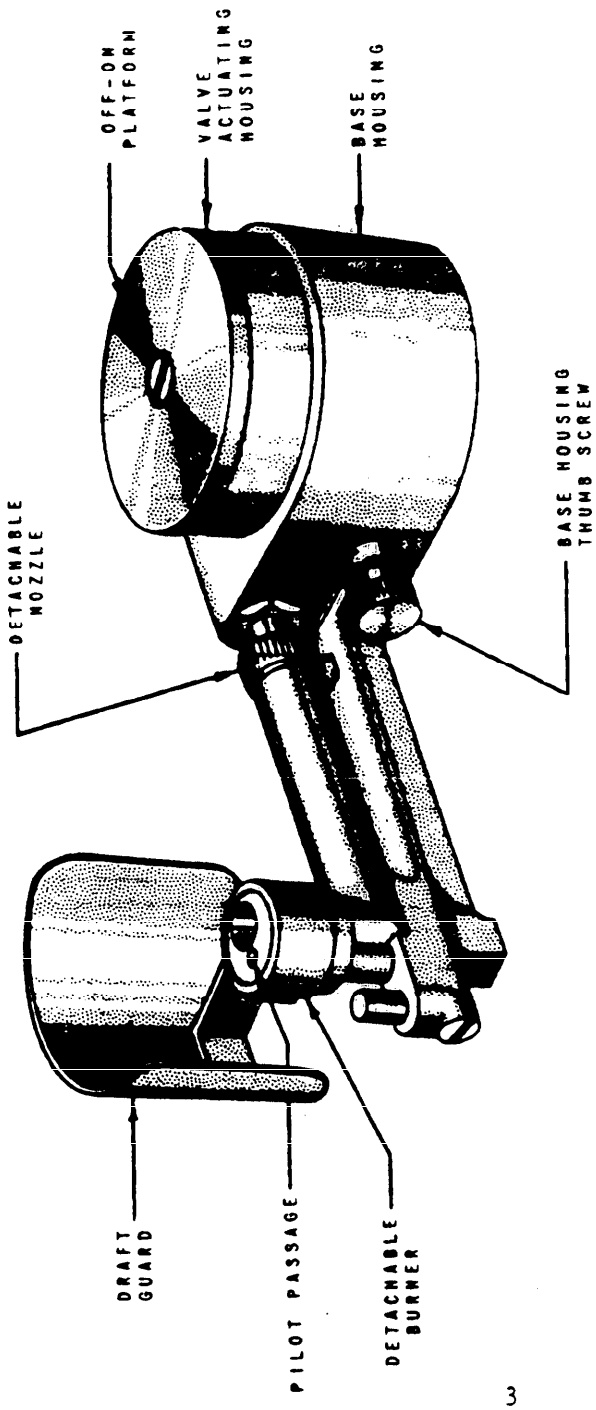


FIGURE 1
GAS BURNER

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either direction shall permit continuous flaming of the main burner. Rotating the platform to its original position shall cause a phosphor bronze spring to disengage the valve actuating mechanism and allow the valve to close off the main burner.

3.2.4 Detachable burner. The detachable burner shall be fabricated of zinc alloy and shall be die cast. The burner shall contain a main gas passage, pilot passage, pilot flame adjustment screw, and air-intake adjustment. Burner shall be attached to the base housing in such a manner that the main gas passage shall be opposite the exterior nozzle orifice of the base housing needle valve, and shall be held securely to the base housing by means of a thumbscrew.

3.2.5 Draft guard. The draft guard shall be fabricated of nickel-plated steel. The guard with the mounting bracket shall be attachable to the burner and shall be rotatable in order to protect the pilot flame and main flame against draft from any direction.

* 3.2.6 Waxing cup. The waxing cup shall be fabricated of nickel-plated steel and shall have the following dimensions: diameter 2-1/2 inches and depth 1/2 inch. Cup shall be mountable and rotatable on a stud provided on the burner, and shall have a loop heat conductor to control the consistency of the wax. The loop heat conductor shall be independently rotatable to permit adjustment for the desired degree of heat transmitted to the waxing cup.

* 3.2.7 Performance. The burner shall perform in accordance with manufacturer's instruction when tested as specified (see 4.3.1).

3.3 Identification marking. Each bunsen burner shall be permanently and legibly marked with the name or registered trademark of the manufacturer.

3.4 Directions. Complete and detailed directions for maintenance and use shall be furnished with each bunsen burner.

3.5 Workmanship. Workmanship shall be first class throughout. Bunsen burners shall be free from defects which detract from their appearance or impair their serviceability.

* 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification to assure supplies and services conform to prescribed requirements.

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4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.2 Quality conformance inspection. The examinations and tests required to assure conformance of those items or lots of items to be offered for acceptance are classified as specified in table I.

4.2.1 Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable quality levels shall be as indicated in table I. Unit of product for examination purposes shall be one burner.

TABLE I. Sampling for examination.

	Inspection level	AQL (defects per hundred units)
For visual examination		
Major defects	II	2.5
Total defects (major and minor defects, combined)	II	4.0

4.2.2 Examination. The burner shall be examined for defects including, but not limited to, those listed in table II.

TABLE II. Classification of defects.

Categories	Defects
Major	
101	Component missing.
102	Burner not securely fastened to base.
103	Burner not free of sharp edges and protrusions.
104	Burner not free of rust, scale, dents, cracks or pits.
105	Draft guard does not rotate 360 degrees.
106	Draft guard and waxing cup not uniformly nickel plated; not free from blisters, chips and peelings.
Minor	
201	Identification marking illegible, incomplete, incorrect or missing.
202	Any component not free of scratches or mar marks.

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4.2.3 Packaging inspection. The inspection of packaging and packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and marking requirements of MIL-STD-129.

4.2.4 Sampling for performance test. Sampling shall be conducted in accordance with MIL-STD-105. Inspection level shall be S-2. Acceptance number shall be zero. Unit of product for test purposes shall be one burner.

4.3 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.3.1 Performance test. The burner shall be connected to a gas supply. The continuous pilot shall be lit to produce a pin point flame. A regulated flame shall be produced instantaneously by the pilot, when the "on-off" platform is depressed. The flame shall be instantaneously and automatically extinguished when the platform is released. The needle valve shall show no evidence of sticking when turned in a clockwise or counterclockwise direction. The draft guard shall show no evidence of sticking or scraping when roated on its axis for a full 360 degrees.

* 5. PACKAGING

5.1 Packaging. Packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Unit package. Each burner shall be packaged in a box of appropriate size and constructed in accordance with PPP-B-676 or PPP-B-636, class domestic. Box shall be designed with suitable interior fittings or shall contain sufficient cushioning material to insure against shifting of, or damage to, the contents. Closure of box shall be as specified in the appendix of the applicable box specification.

5.1.2 Level C. Burners shall be individually packaged in a manner that will afford adequate protection against corrosion, deterioration and physical damage during shipment from supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A. Forty eight (48) burners shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant. Closure and strapping shall be as specified in the appendix of the applicable box specification.

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5.2.1.1 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes. Each fiberboard box shall be waterproofed as specified in the appendix of PPP-B-636.

5.2.2 Level B. Forty eight (48) burners shall be packaged in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification.

5.2.3 Level C. The packaged burners shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. These packs shall conform to the applicable carriers' rules and regulations.

5.3 Marking. Each unit package and exterior container shall be marked as specified in MIL-STD-129.

6. NOTES

* 6.1 Intended use. The burner shall be suitable for use in the dental prosthetic laboratory.

* 6.2 Ordering data. Procurement documents should specify the following.

(a) Title, number and ate of this specification.
(b) National Stock Number (NSN).
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

* 6.3 The automatic bunsen burner covered by this specification is listed in the Federal Supply Catalog as follows.

National Stock Number	Item Identification
6640-00-510-6075	BURNER, GAS, LABORATORY, Bunsen, Automatic

* 6.4 This specification does not cover all types, classes, grades or sizes of th commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the type normally procured to meet requirements.

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* 6.6 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of marginal notations and relationship to the last previous issue.

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-B-36274A BURNER, GAS, LABORATORY, BUNSEN, AUTOMATIC

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